The RoboCup Humanoid Soccer League
Fitness Tests

RoboCup Technical Committee

(2026 fitness tests, as of 2026-06-24)

Questions or comments on the fitness test rules should be submitted via `https://github.com/RoboCup-HumanoidSoccerLeague/HSL-Rules/issues`, to the `#rule-book` channel on the RoboCup Official Discord server, or by mail to `rc-hsl-tc@lists.robocup.org`.

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1 Introduction

Starting in 2026, a fitness test will be introduced to track and compare team performances across key skills over multiple years. These fitness tests track the development of the league and are used to make informed decisions on rule changes such as field sizes or game duration. They further aim to motivate teams to improve in specific areas valuable to humanoid robotic soccer and to compile a record of the top-performing components and skills. By highlighting these individual skills, the league encourages focused development in areas that contribute to overall team performance, and creates a showcase of the best technical abilities in the league.

The 2026 fitness tests will be a trial run to ensure feasibility, and as such will not have any impact on team scores and will not award any certificates or trophies. Failing to complete these fitness tests will not have any negative repercussions. However, all teams are required to show up to their assigned slots and must declare if they are unable to complete any of the challenges.

1.1 Challenge Rules

The fitness tests are conducted as part of the robot inspection and consist of three individual challenges as described below. Each challenge can be attempted up to three times and should be attempted at least once. If a robot is unable to complete any of the challenges, the team may declare this and thus skip the challenge. A skipped challenge is marked as ”not attempted”. If a challenge is attempted multiple times, the best result is counted.

Challenges require a single robot. If a team has multiple different robot platforms, they may ask for permission to make a second attempt at the fitness test with another robot platform, but are not required to do so. The OC may reject this request if the schedule does not allow for it. Teams are not allowed to use multiple different robot platforms in the same fitness test attempt.

1.2 Setup

The fitness tests are conducted on a small division field. Additional required materials include a stopwatch, a tape measure and one ball of each division.

1.3 Results

The results of the fitness tests will be recorded and published on the Humanoid Soccer League website (hsl.robocup.org). Results will be kept indefinitely to allow analysis of the development of the league over time.

Unless otherwise specified, all results are recorded in SI base units with 2 decimals of accuracy.

2 Fastest Walk Test

This fitness test measures the fastest straight autonomous walk from one side of the field to the other.

2.1 Setup

The competing robot is placed 50cm behind the starting goal line at the height of the intersection with the penalty area, facing the opposite goal line. The setup is illustrated in Figure 1.

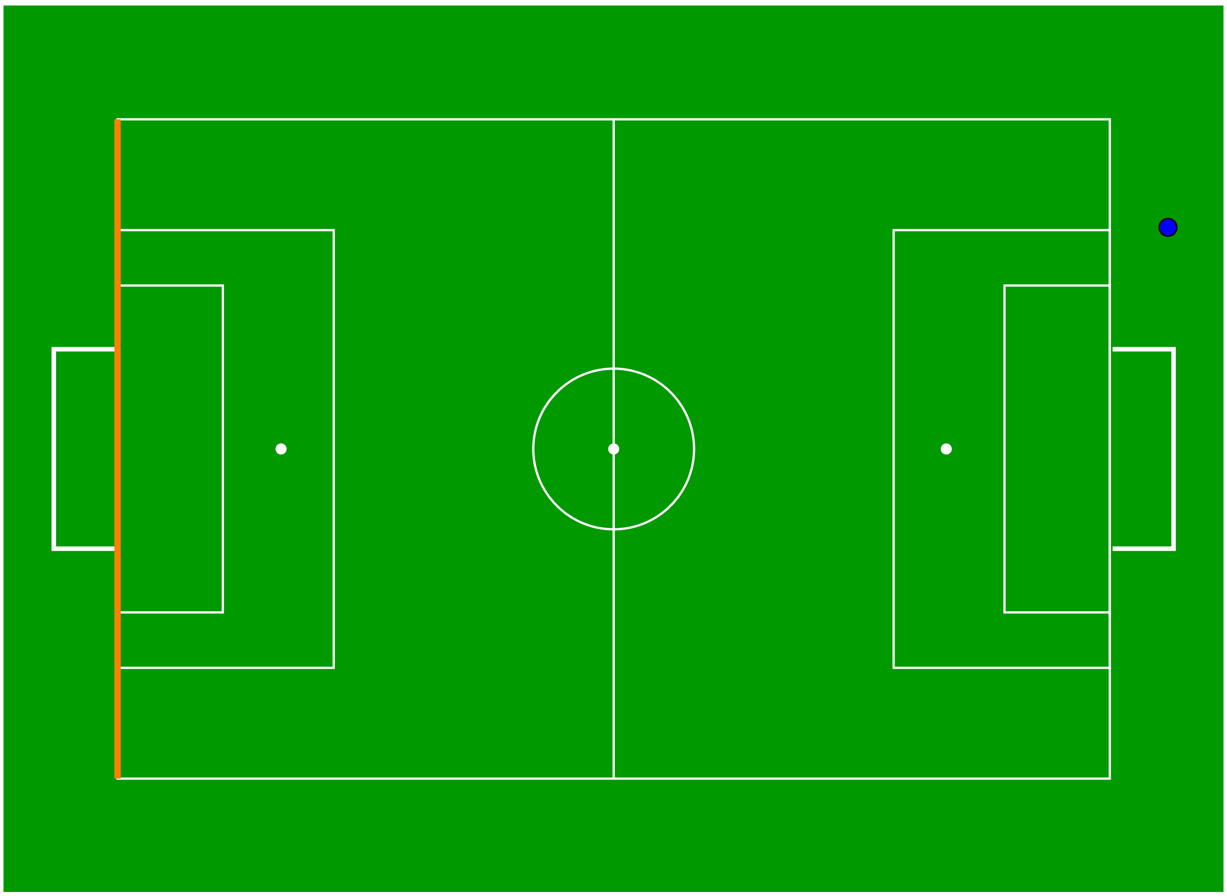

Figure 1: Fastest Walk Fitness Test (S-Field): the start point is shown in blue and the finishing goal line is highlighted in orange.

2.2 Challenge Execution

In each attempt, the active robot must walk autonomously in a straight run from the starting side of the field to the opposite goal line. Code changes are not allowed during an attempt, but are allowed between attempts.

The run ends when the robot entirely crosses the opposite goal line. The attempt also ends if the robot falls and is unable to get up within 20 seconds or if the total duration of the attempt exceeds one minute (as determined by "Scoring" below).

2.3 Scoring

The head referee starts the timer once the robot entirely crosses the starting goal line and stops the timer once the target goal line is entirely crossed. The lowest time across all attempts is recorded.

If the robot is unable to reach the target goal line, the distance traveled is recorded instead, measured as the projection onto the touchline of the segment from the point where the robot crossed the starting goal line to the closest point of robot's final position.

3 Kick Accuracy Test

This fitness test measures the accuracy with which a robot can kick an HSL competition soccer ball toward specified targets.

3.1 Setup

This challenge additionally requires a ball of each division. Teams may choose which division's ball they want to use for the challenge. The ball is placed on the center of the edge of the goal area. The competing robot is placed in the center of the goal line facing the ball.

3.2 Challenge Execution

Each attempt consists of kicking toward each of the three targets (illustrated in Figure 2):

- The center mark in the center circle
- The left corner on the opposite side of the field

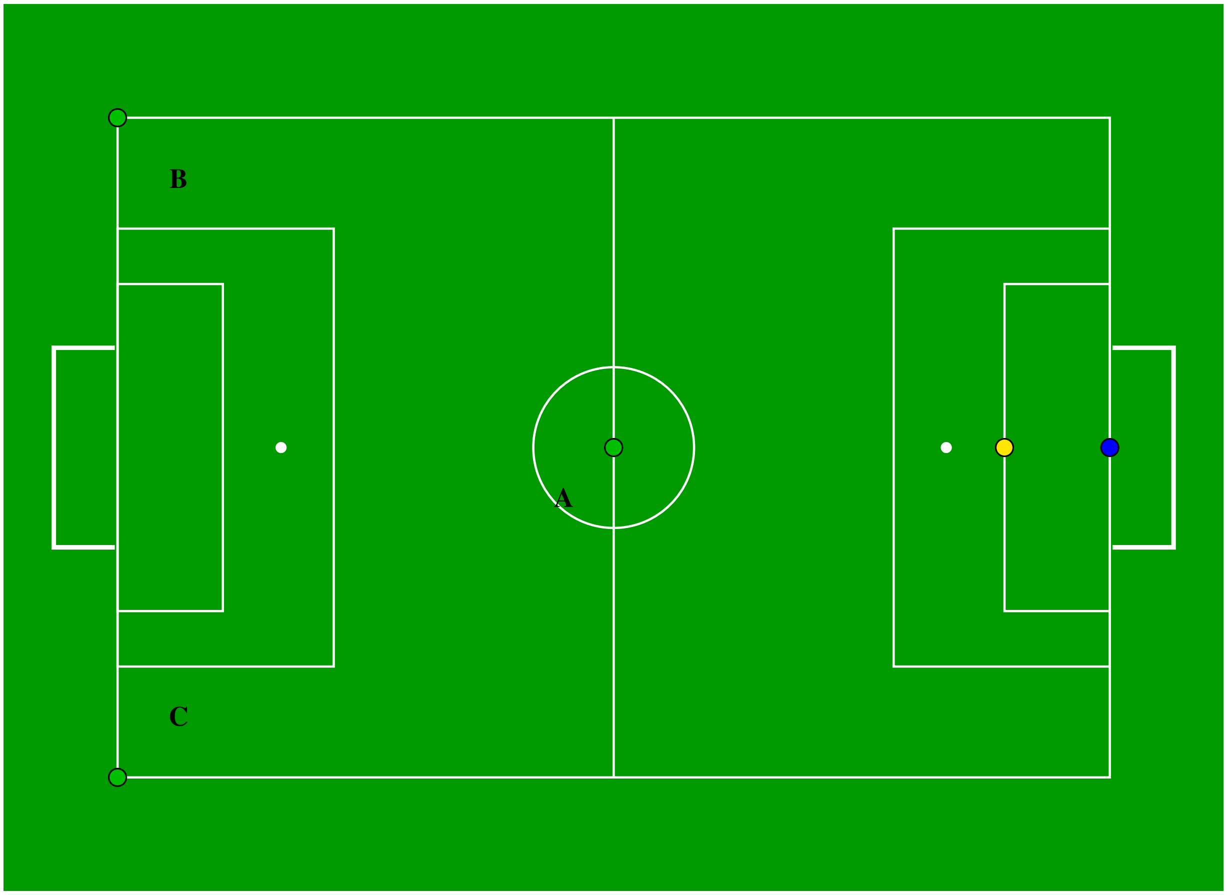

Figure 2: Kick Accuracy Fitness Test (S-Field): robot is shown in blue, targets are shown in green, and ball placement is shown in yellow.

- The right corner on the opposite side of the field

The robot is placed at its starting position before each kick.

In each attempt, for each kick, participating teams must start the robot behavior of their competing robot via a single button press. The robot then has 60 s to kick the ball towards the current target. The competing robot must not touch the ball a second time after the ball has clearly moved, otherwise the attempt ends immediately without scoring and the robot continues with the next target. Once the ball has come to a complete stop, the distance from the ball to the target is measured.

Code changes are allowed when changing targets.

3.3 Scoring

After all attempts, the lowest distance achieved for each target is recorded. Each target is scored independently.

4 Recovery from Fall Fitness Test

This fitness test measures how quickly a robot can recover and resume normal operation after falling forward or backward.

An attempt of this test consists of two separate get-up trials: one starting from the prone (face down) position and one from the supine (face up) position.

4.1 Setup

The robot is placed already on the ground in the center of the field in the assigned position (flat on the front or back).

4.2 Challenge Execution

After being placed lying on the ground, the robot must autonomously recover and stand up.

For each attempt, for each starting position, teams start the robot recovery behavior via a single button press. Code changes are allowed when changing starting positions.

4.3 Scoring

The head referee starts timing at the button press, and stops the timer when the robot is standing on both feet. The robot must remain stable for at least three seconds after reaching its standing position for the time to be counted. The fastest recovery time across all attempts for each starting position is recorded. Each starting position is scored independently.